

1. Suppose $A = \{1\}$, $B = \{2\}$, $C = \{3\}$, and $D = \{4\}$. Find the following:

$$(A \times B) \cup (C \times D)$$

$$A \times B = \{\{1, 2\}\}$$

$$C \times D = \{\{3, 4\}\}$$

$$(A \times B) \cup (C \times D) = \{\{1, 2\}, \{3, 4\}\}$$

$$(A \cup C) \times (B \cup D)$$

$$A \cup C = \{1, 3\}$$

$$B \cup D = \{2, 4\}$$

$$(A \cup C) \times (B \cup D) = \{\{1, 2\}, \{1, 4\}, \{3, 2\}, \{3, 4\}\}$$

2. Suppose A , B , C , and D are any sets. Based on your answer above, how do you think $(A \times B) \cup (C \times D)$ and $(A \cup C) \times (B \cup D)$ are related? Is one a subset of the other? Are they equal? Try to prove your answer.

From what I can see, $(A \times B) \cup (C \times D)$
is a subset of $(A \cup C) \times (B \cup D)$.

The proof is $\{1, 2\}$ and $\{3, 4\}$ are in
 $(A \cup C) \times (B \cup D)$.